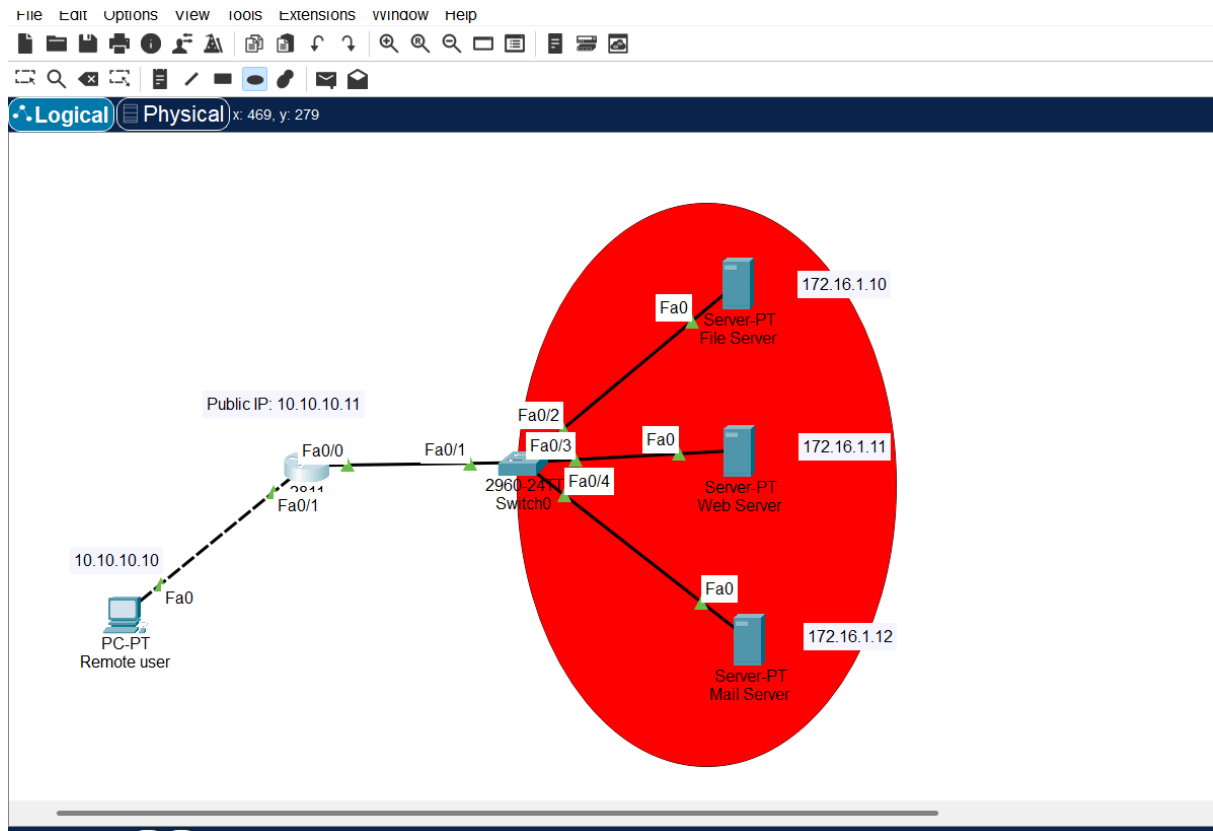
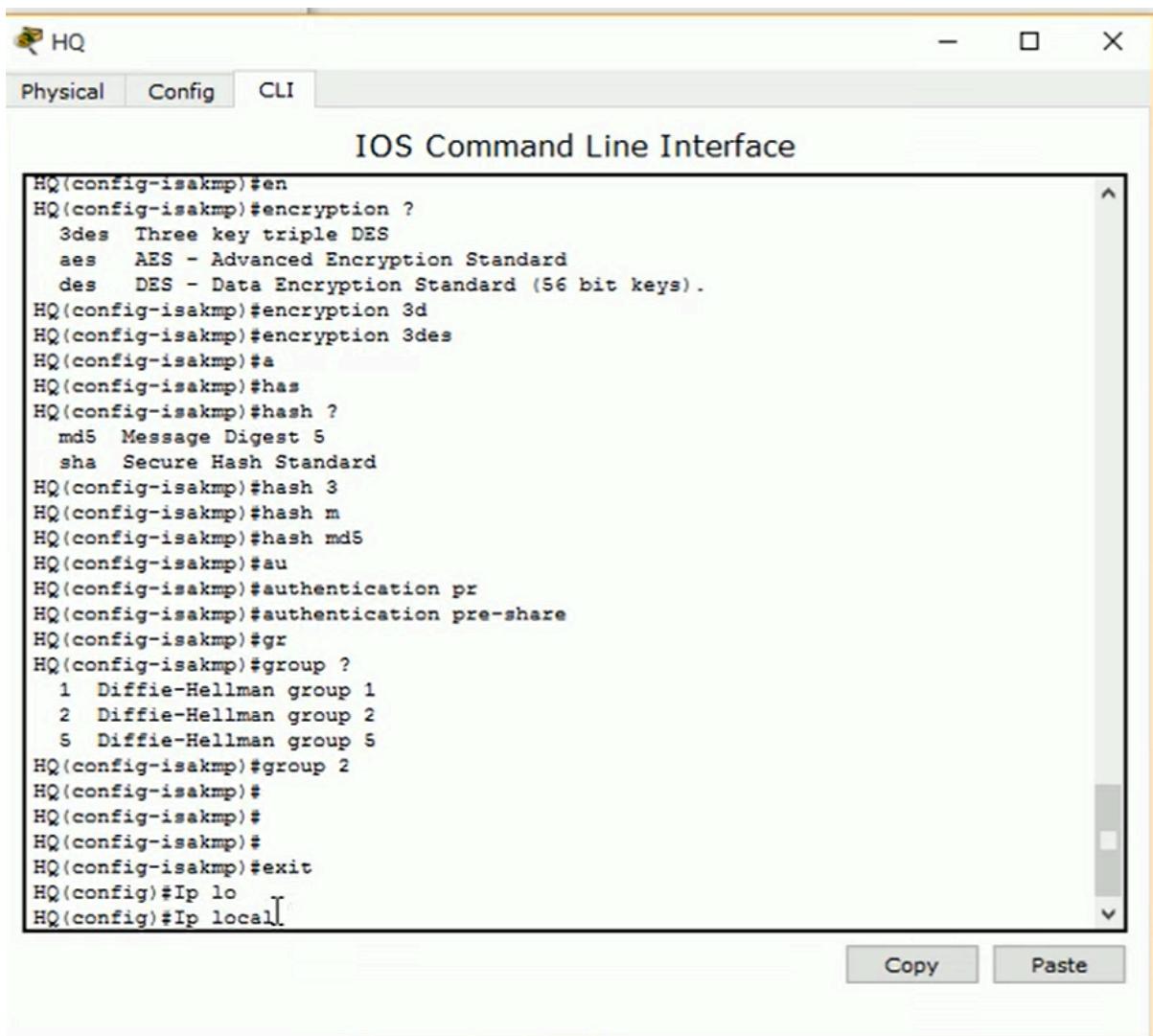


Lab 10: VPN and IPSec:





IOS Command Line Interface

```
HQ(config)#Ip local pool VPNPOOL 192.168.1.1 192.168.1.50
HQ(config)#
HQ(config)#
HQ(config)#cry
HQ(config)#crypto is
HQ(config)#crypto isakmp cl
HQ(config)#crypto isakmp client con
HQ(config)#crypto isakmp client configuration gr
HQ(config)#crypto isakmp client configuration group ?
WORD group name
HQ(config)#crypto isakmp client configuration group cisco
HQ(config-isakmp-group)#ke
HQ(config-isakmp-group)#key cisco123
HQ(config-isakmp-group)#Pool
HQ(config-isakmp-group)#Pool VPNPOOL
HQ(config-isakmp-group)#
HQ(config-isakmp-group)#
HQ(config-isakmp-group)#exit
HQ(config)#cry
HQ(config)#crypto ip
HQ(config)#crypto ipsec tr
HQ(config)#crypto ipsec transform-set ?
WORD Transform set tag
HQ(config)#crypto ipsec transform-set set1 esp
HQ(config)#crypto ipsec transform-set set1 esp-3
HQ(config)#crypto ipsec transform-set set1 esp-3des esp
HQ(config)#crypto ipsec transform-set set1 esp-3des esp-m
HQ(config)#crypto ipsec transform-set set1 esp-3des esp-md5-hmac
HQ(config)#
HQ(config)#
HQ(config)#
HQ(config)#cry
HQ(config)#crypto dy
```

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IOS Command Line Interface

```
WORD Dynamic crypto map template tag
HQ(config)#crypto dynamic-map map1 ?
<1-65535> Sequence to insert into dynamic-map entry
HQ(config)#crypto dynamic-map map1 10
HQ(config-crypto-map)#se
HQ(config-crypto-map)#set tr
HQ(config-crypto-map)#set transform-set set1
HQ(config-crypto-map)#re
HQ(config-crypto-map)#reverse-route rou
HQ(config-crypto-map)#reverse-route
HQ(config-crypto-map)#
HQ(config-crypto-map)#
HQ(config-crypto-map)#
HQ(config-crypto-map)#
HQ(config-crypto-map)#exit
HQ(config)#
HQ(config)#cry
HQ(config)#crypto m
HQ(config)#crypto map map1 cli
HQ(config)#crypto map map1 client con
HQ(config)#crypto map map1 client configuration ad
HQ(config)#crypto map map1 client configuration address re
HQ(config)#crypto map map1 client configuration address respond
HQ(config)#cry
HQ(config)#crypto m
HQ(config)#crypto map map1 cli
HQ(config)#crypto map map1 client aut
HQ(config)#crypto map map1 client authentication li
HQ(config)#crypto map map1 client authentication list abc1
HQ(config)#cry
HQ(config)#crypto m
HQ(config)#crypto map is
HQ(config)#crypto map is
```

Copy

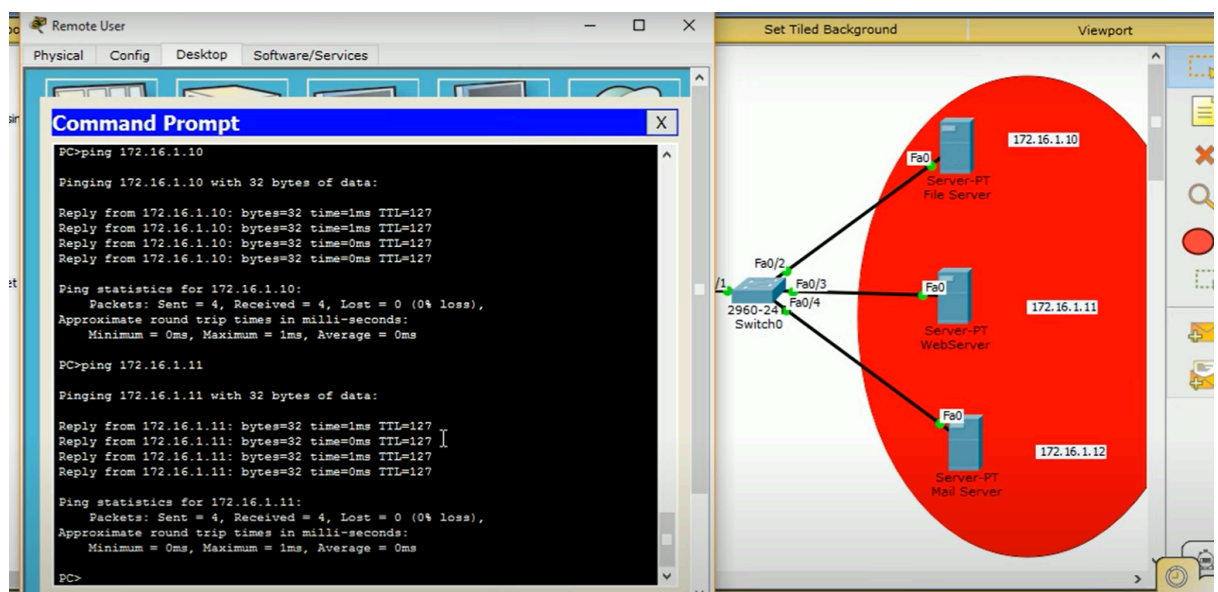
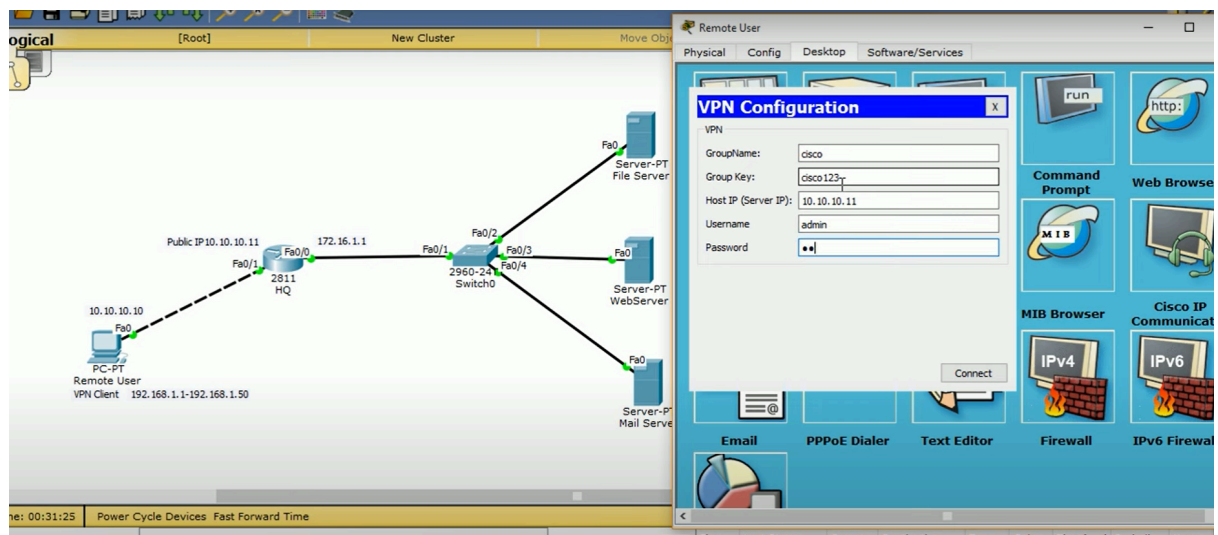
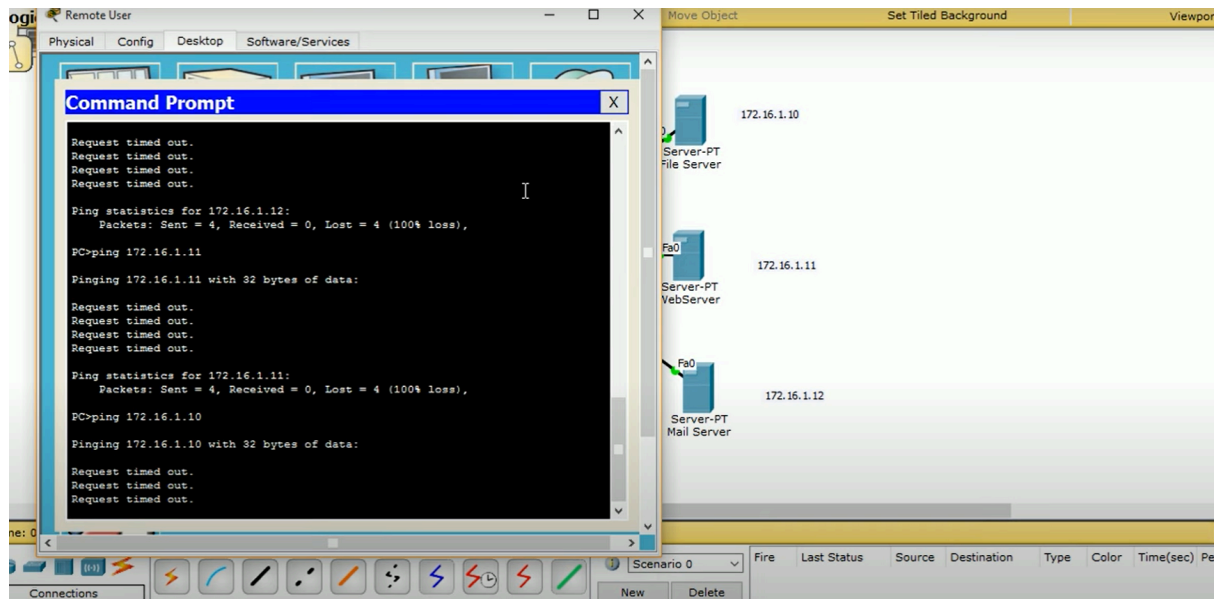
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IOS Command Line Interface

```
HQ(config)#crypto map map1 cli
HQ(config)#crypto map map1 client con
HQ(config)#crypto map map1 client configuration ad
HQ(config)#crypto map map1 client configuration address re
HQ(config)#crypto map map1 client configuration address respond
HQ(config)#cry
HQ(config)#crypto m
HQ(config)#crypto map map1 cli
HQ(config)#crypto map map1 client aut
HQ(config)#crypto map map1 client authentication li
HQ(config)#crypto map map1 client authentication list abc1
HQ(config)#cry
HQ(config)#crypto m
HQ(config)#crypto map is
HQ(config)#crypto map isaut
HQ(config)#crypto map isa
HQ(config)#crypto map map1 is
HQ(config)#crypto map map1 isakmp aut
HQ(config)#crypto map map1 isakmp authorization li
HQ(config)#crypto map map1 isakmp authorization list abc2
HQ(config)#cyr
HQ(config)#cru
HQ(config)#cry
HQ(config)#crypto m
HQ(config)#crypto map map1 10 ip
HQ(config)#crypto map map1 10 ipsec-isakmp dy
HQ(config)#crypto map map1 10 ipsec-isakmp dynamic map1
HQ(config)#
HQ(config)#
HQ(config)#int f0/1
HQ(config-if)#cry
HQ(config-if)#crypto ma
HQ(config-if)#crypto map ma
```

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Explanation:

In this lab, we configured a VPN Remote Access solution with IPSec on a Cisco router using Cisco Packet Tracer. We started by assigning IP addresses to the router's interfaces to ensure connectivity between the internal network, the public network, and the remote user. We then created local login credentials so the remote user could securely authenticate. Next, we configured IKE Phase 1 by setting encryption, hashing, and key exchange parameters, establishing a secure tunnel initiation. We defined a VPN IP address pool to allocate IPs to connecting clients, and created a VPN group with a group name and pre-shared key for authentication. Then, we configured the IPSec transform set (IKE Phase 2) to specify how the actual data traffic would be encrypted and authenticated. We set up a dynamic crypto map to allow the router to dynamically handle VPN connections from various remote clients. After applying the crypto map to the public-facing router interface, we tested the VPN connection from a remote PC by filling in the VPN client settings and connecting to the router. Finally, we verified the VPN tunnel status and ensured that data was encrypted by using router commands and a packet sniffer, confirming that the setup successfully secured remote access to the company's internal servers.